

Applied Statistics: Time Series

Robin Delabays

September 11, 2026

Table of contents

Table of content	4
Fall 2026	4
 I Week 1	 5
Week 1: Course introduction	6
Course organization	6
Day and time	6
Online material	6
Exams	6
Course wiki	6
Course (approximate) schedule	7
Special needs	7
Working practice	8
LLMs...	8
Contact	8
 Week 1: Introduction to time series	 9
Definition of time series	9
Examples of time series	9
Acquisition	10
Time step	10
Recording time	10
Resampling	10
Representing	11
Noise	11
Hands on	11
 II Week 2	 12
 III Week 3	 13

IV Week 4	14
V Week 5	15
VI Week 6	16
VII Week 7	17
VIIIWeek 8	18
IX Week 9	19
X Week 10	20
XI Week 11	21
XII Week 12	22
XIIIWeek 13	23
XIVWeek 14	24
XV Week 15	25

Table of content

This page contains the material for the class *Applied Statistics: Time Series*.

Fall 2026

- [September 15, 2026 \(Course intro\)](#)
- [September 15, 2026](#)
- September 22, 2026
- September 29, 2026
- October 6, 2026
- October 8, 2026
- October 15, 2026
- October 20, 2026
- November 3, 2026
- November 9, 2026
- November 10, 2026
- November 17, 2026
- November 19, 2026
- December 10, 2026
- December 17, 2026
- January 19, 2027

Part I

Week 1

Week 1: Course introduction

[Slides](#) [Slides PDF](#)

Course organization

Day and time

Variable throughout the semester

Online material

All material will be made available from [this webpage](#).

Exams

First exam on November 10, 2026: Brief exam (most likely QCM), approx. 45 minutes.

Second exam on January 19, 2027: Longer exam with hands on, approx. 90 minutes.

Course wiki

A course wiki will be set up. The idea is that you create your own list of important concepts. You fill it, I check it.

Course (approximate) schedule

15.9.26: Intro to the course and intro to time series

22.9.26: Recap of statistics

29.9.26: Advanced basics

6.10.26: *Lab on time series basics*

8.10.26: Recap on vectorial geometry

15.10.26: Fourier 1/2

20.10.26: Fourier 2/2

3.11.26: *Lab on Fourier*

9.11.26: Recap and mock exam

10.11.26: CC and correction

17.11.26: Autoregressive models

19.11.26: Moving average (recap) and ARMA models

10.12.26: ARMA fitting

17.12.26: *Lab on ARMA*

19.1.27: Final exam

Special needs

Some of you may need some specific arrangements for the courses and/or the exams.

1. My first suggestion is to make it official by talking to your *Responsable de filière* (Mme Birgit Sievert).
2. In case you need indeptendent help, you can reach out to our mediator: Laurence Laffargue-Rieder, laurence.laffargue@hevs.ch
3. Finally, I strongly encourage you to tell me what I can do to help.

Additionnal reading: [Diversité et réussite dans l'enseignement supérieur](#)

Working practice

You are now in third year and I am sure that you know how to learn. I like to recap some (of my) good practices:

- **Taking notes** helps you stay active and attentive to the class.
- **Taking breaks** helps you study longer. More than two hours of studying in a row is counterproductive.
- Remember that studying is a **social activity**. Of course, you are developing new skills for your future job, but you are also making new friendships that may last for long. So again, take some breaks.
- For the same amount of time dedicated to the course, spreading it over a long period is much more efficient than using it all at the same time. **Regularity** is the key!

LLMs...

I will not tell you not to use it, but as everyone else, I will tell you to use it with care. LLMs are very strong tools for learning, but they can also dramatically deteriorate your work.

Use LLMs as sparring partners. Talk with it, challenge it. Make sure that you understand how and why it gave an answer.

If the LLM only gives you the answer, then you have learned nothing.

Cheating: We have had some strong suspicions of cheating using LLMs in the past. We will be extra careful from now on.

Contact

- Robin Delabays
- Offices: 21.N203 or 23.N403
- E-mail: robin.delabays@hevs.ch
- Phone: +41 58 606 86 09

Week 1: Introduction to time series

[Slides](#) [Slides PDF](#)

Definition of time series

According to you, what is a time series?

Two main ingredients: 1. Series: A “thing”, usually measurable, i.e., a number. 1. Time: This number changes over time.

Formally, a time series is a sequence of values each indexed by a **time step**:

$$X = (x_{t_1}, x_{t_2}, \dots, x_{t_N}),$$

where x_t is the value (usually a real number) of the time series at time t and $t_1, t_2, \dots$ are the time steps. Each element x_{t_i} of a time series is called a **sample**.

In general, there is no constraints on the time steps. But for this course, we will always consider times steps are evenly spaced. Therefore, we will skip the t in the indexing and just number the components of the time series:

$$X = (x_1, x_2, \dots, x_N).$$

Note that we then have to specify the time of first recording and the **sampling rate**, i.e., the time between two consecutive samples.

TODO: Add a figure here.

Examples of time series

- Any meteorological quantity, measured over time (temperature, pressure, wind speed,...)
- Stock market prices
- Any quantity measured inside a bioreactor (temperature, pH,...)
- Any health constant measured on a patient (temperature, heart rate, blood pressure,...)

TODO: Figures?

Acquisition

Time step

The acquisition of measurements requires the recording device to “read” the phenomenon and then “write” the recorded value. By nature, a time series cannot be continuous, because there is necessarily a gap between two consecutive values. Depending on the context, however, this gap can be made extremely small.

In practice, the duration of a time step, or sampling rate, should be made as small as needed for the purpose, but as large as possible to spare memory and computation.

Example: *Recording sounds requires to sample at a frequency of approximately 20 kHz (largest audible frequency). Actually, we need to record at twice this frequency because of aliasing (see later, Nyquist’s Theorem).*

In this course, the time step of a time series will be denoted as δ_t , measured in seconds.

Recording time

Depending on the context, there is also a minimal recording time required in order to observe certain phenomena.

Example: *In sound recording, if one does not record for more than 0.05 seconds, some audible sounds may not be recorded, because the lower bound of the audible spectrum is 20 Hz, i.e., 20 oscillations per second.*

In this course, the duration of a time series will be denoted T , measured in seconds.

Resampling

Sometimes, the given time step of a time series is not appropriate for our purpose.

Example: *I want to study the impact of the season on temperature. If the time step of my time series is at 15 minutes, this is too fine.*

Given a fine time series, coarse graining is easy, but comes with a question. For each coarser time step, I have multiple values measured. Which one(?) should I use?

TODO: Have illustrating figures

Representing

The first step in the analysis of a time series is visual inspection.

Make sure to use an appropriate plotting tool. You need to be able to rescale, zoom in, zoom out of you time series.

I recommend pyplot from matplotlib, which is a powerful plotting library.

Noise

Example: *Measure the height of someone in millimeters.*

Any data acquisition is imperfect. Deviations of the measurements from the “real” system’s values are called **noise**. Noise is usually added (or subtracted) from the measured values in the acquisition process.

One goal in time series analysis will be to mitigate the impact of noise.

In general, we will assume that the noise has zero mean. Therefore, the measured signal remains around the real signal.

Hands on

TODO

Part II

Week 2

Part III

Week 3

Part IV

Week 4

Part V

Week 5

Part VI

Week 6

Part VII

Week 7

Part VIII

Week 8

Part IX

Week 9

Part X

Week 10

Part XI

Week 11

Part XII

Week 12

Part XIII

Week 13

Part XIV

Week 14

Part XV

Week 15